EMS
Entacrest MFB Solution

User guide

How to install and set up EMS, step by step, and how to use every feature: customers, savings, thrift, loans, transfers, field collections, approvals and reports.

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An Entacrest Nexus product

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CHAPTER 1

What EMS does

EMS runs the day-to-day work of a microfinance bank, cooperative or thrift society, from the counter to the field to the MD's desk.

Area	What you can do
Customers and accounts	Register customers with their ID, photo and next of kin. Open savings, target savings, thrift and loan accounts. Print statements.
Counter	Post deposits and withdrawals, with automatic withdrawal charges. Large withdrawals wait for a manager's approval.
Transfers	Move money between a customer's own accounts, or to another customer. Paying into a loan counts as a repayment.
Thrift	Daily, weekly or monthly contribution plans with commission, cycles and payouts.
Field app	Account officers collect money in the market on a phone, even offline, and close their day. The accountant counts their cash.
Loans	Loan products with monthly or overall interest, processing fees and daily late fees. Applications, approval, payout, repayment schedules, top-ups and extensions.
Control	Approvals for sensitive actions, reversals, edits by senior staff only, and a full audit log of who did what.
Reports	Staff and branch performance, income, loan portfolio, defaulters, dormant accounts and more, exported to PDF or CSV.

Who uses EMS

Everyone signs in with their own email and password, plus a 6-digit code sent to their phone. What each person sees depends on their **role** and the **permissions** the MD gives them. See Chapter 20 for a summary.

Two ways to use it

- **Desktop app (Windows).** Install it on your office computer. This guide mostly describes the desktop app.
- **Web.** Your organisation can also host EMS online and staff open it in a browser. The screens are the same.

CHAPTER 2

Before you start

Have these ready. The whole set-up takes about 15 minutes.

1. **A Windows 10 or 11 computer** for the head office. It will hold your data, so use a reliable machine.
2. **The installer file**, `EMS-Setup-0.1.0.exe`, from Entacrest.
3. **Your licence code** from Entacrest. It is a long block of text starting with `EMS-`. Your licence sets how many branches you can run and when it must be renewed.
4. **Your business details:** name, head office address, phone number.
5. **The MD's details:** name, work email, mobile number and a strong password (at least 8 characters with a capital letter, a lowercase letter and a number).
6. **Your branch list**, each with a two-digit code (01, 02…) and the market areas officers collect from.
7. **Your staff list**: names, emails, mobile numbers, roles and branches.

> **Want to look around first?**
> The desktop app includes a demo workspace full of sample data, separate from your real data. See Chapter 17.

CHAPTER 3 · STEP BY STEP

Install EMS

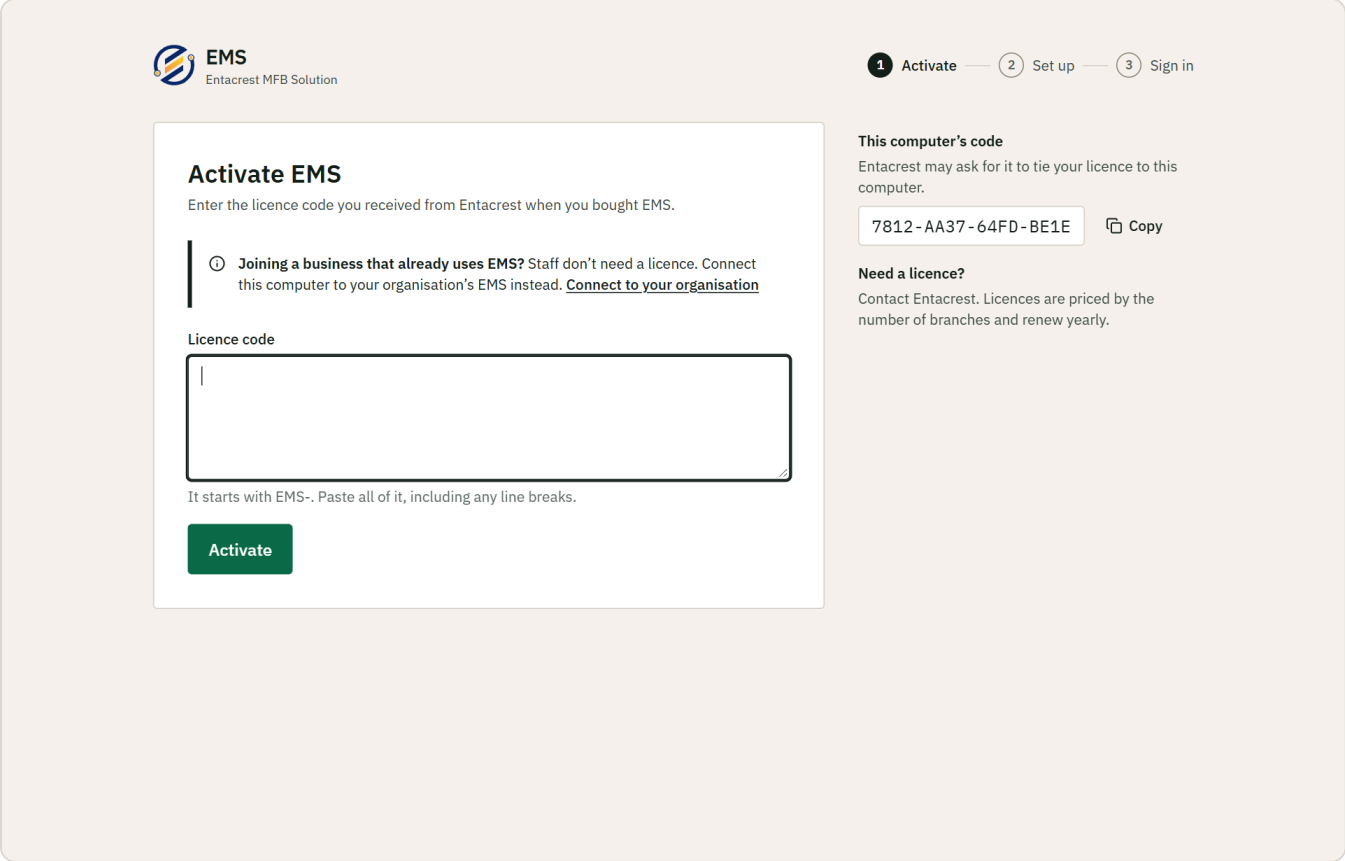
1. Copy `EMS-Setup-0.1.0.exe` to the computer and double-click it.
2. If Windows shows **“Windows protected your PC”**, click **“More info”**, then **“Run anyway”**. This appears because the app is new to Windows. It is safe.
3. Follow the installer. Keep the suggested folder unless you have a reason to change it, then click **“Install”**.
4. When it finishes, EMS opens. Next time, open it from the **EMS** shortcut on the desktop or the Start menu.

> **Where your data lives**
>
> By default EMS keeps its database on this computer, in `%APPDATA%\EMS\data`. Back it up regularly (Chapter 18). To use EMS on several computers, see Chapter 7.

CHAPTER 4 · STEP BY STEP

Activate your licence

The first screen asks for your licence. Only the head office computer needs one. Staff on other computers connect to it instead (Chapter 7).



The Activate screen. "This computer's code" is on the right.

1. Open the licence code Entacrest sent you. Copy all of it, from `EMS-` to the end, including any line breaks.
2. Paste it into the **"Licence code"** box.
3. Click **"Activate"**.

> **Entacrest asked for "this computer's code"?**
> Click **"Copy"** next to it and send it to them. Some licences are tied to one computer.

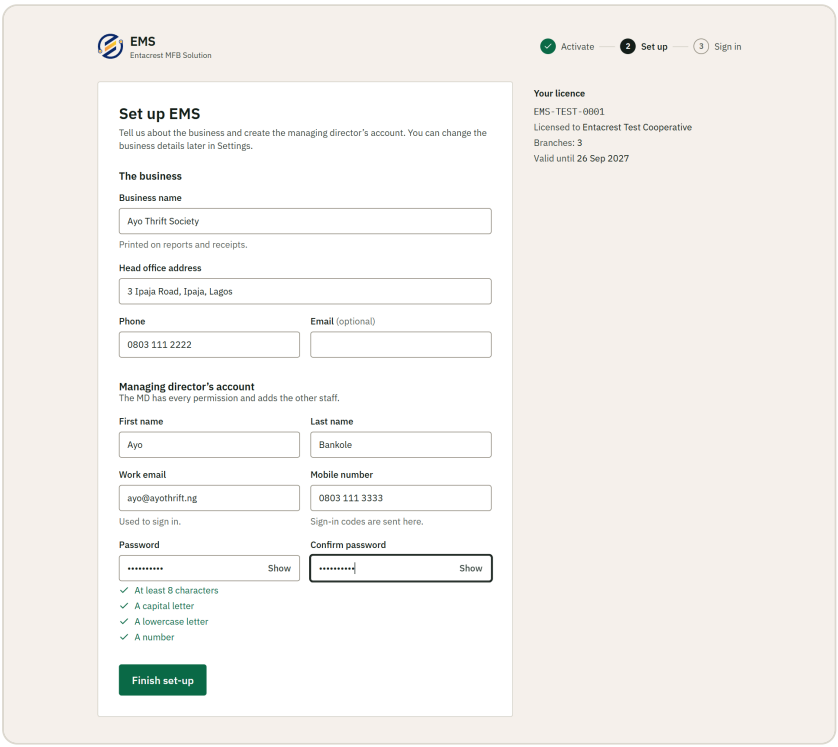
> **Staff installing on their own computer?**
> Don't enter a licence. Click **"Connect to your organisation"** in the grey box instead (Chapter 7).

CHAPTER 5 · STEP BY STEP

Set up your business and sign in

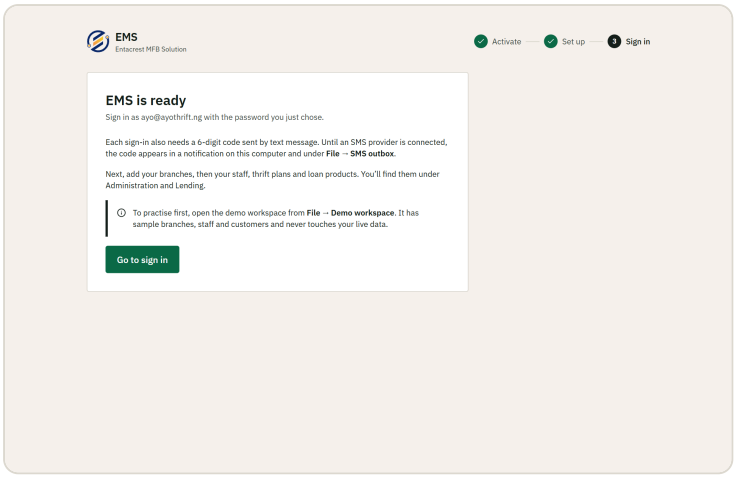
A. Create the business and the MD’s account

1. Under **The business**, enter the business name (printed on reports and receipts), head office address and phone. Email is optional.
2. Under **Managing director’s account**, enter the MD’s first and last name, work email and mobile number. Sign-in codes go to this number.
3. Choose a password and type it again. The ticks turn green when it is strong enough.
4. Click **“Finish set-up”**.



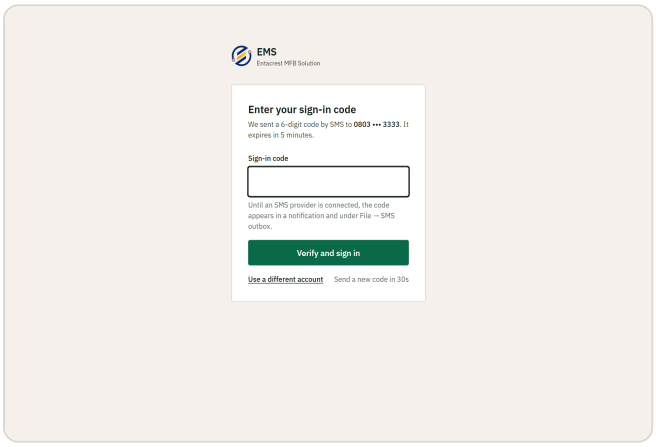
Set up EMS. You can change the business details later in Settings.

B. Sign in for the first time

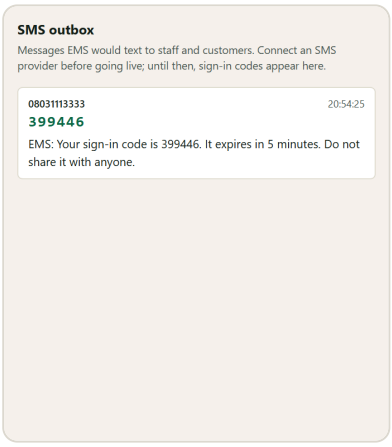


Set-up complete.

1. Click **“Go to sign in”**.
2. Enter the MD’s work email and password, then press **Enter**.
3. EMS asks for a 6-digit code sent by SMS. **Until you connect an SMS provider, the code is not texted.** Open **File → SMS outbox** (or press **Ctrl+3**) to see it.
4. Type the code and press **Enter**. The code expires after 5 minutes.



Enter the sign-in code.



File → SMS outbox shows messages until SMS is connected.

> **Before you go live, connect SMS**
>
> Customers get receipts, and staff get sign-in codes, by SMS. Ask Entacrest to connect Termii or Africa’s Talking for you. Until then, only this computer can see the outbox.

CHAPTER 6 · STEP BY STEP

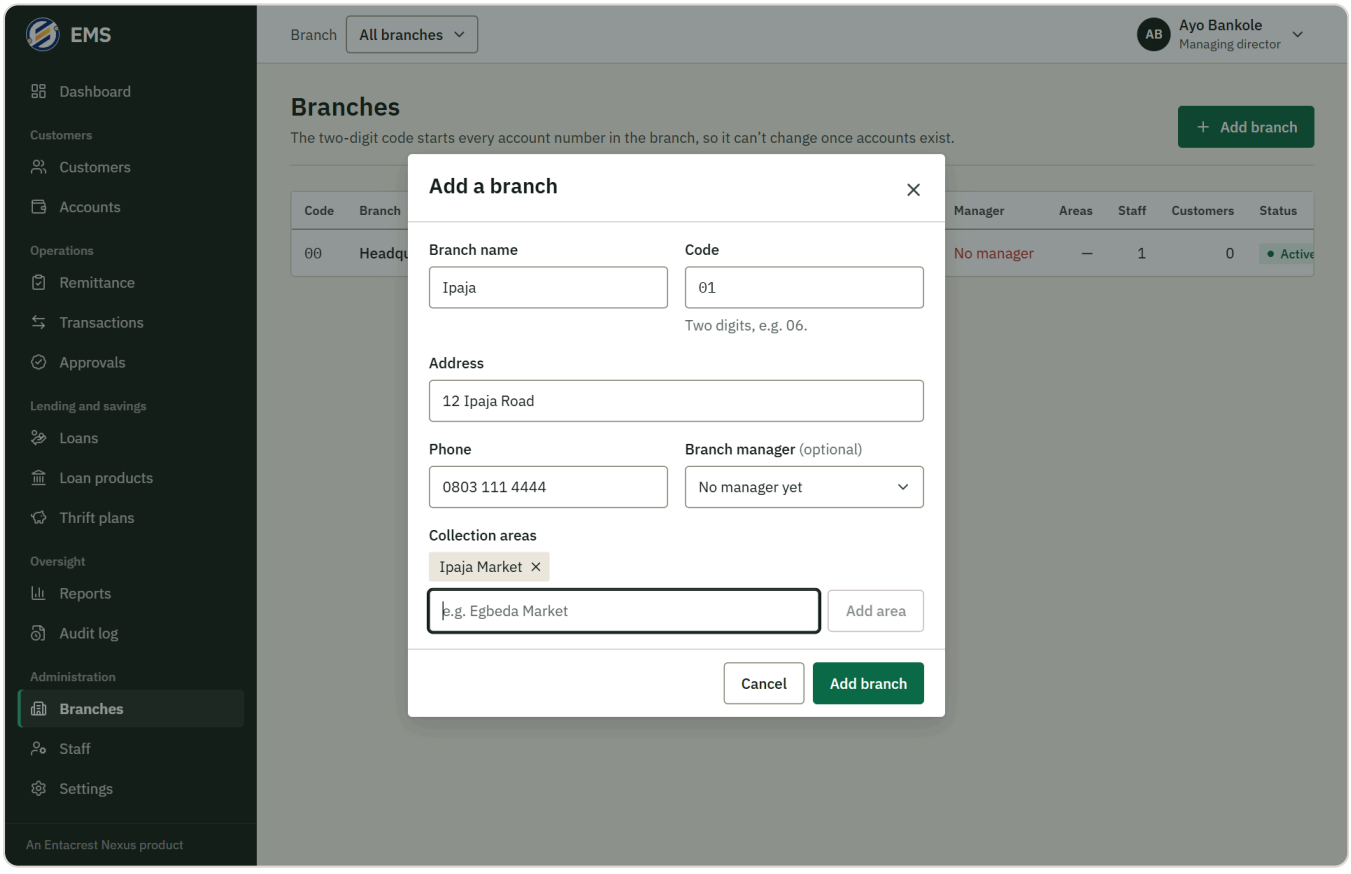
Your first week: set-up checklist

Do these in order. The MD (or an administrator) can do all of them.

Step 1 · Add your branches

A **Headquarters** branch (code 00) already exists. It is where the MD and general manager sit.

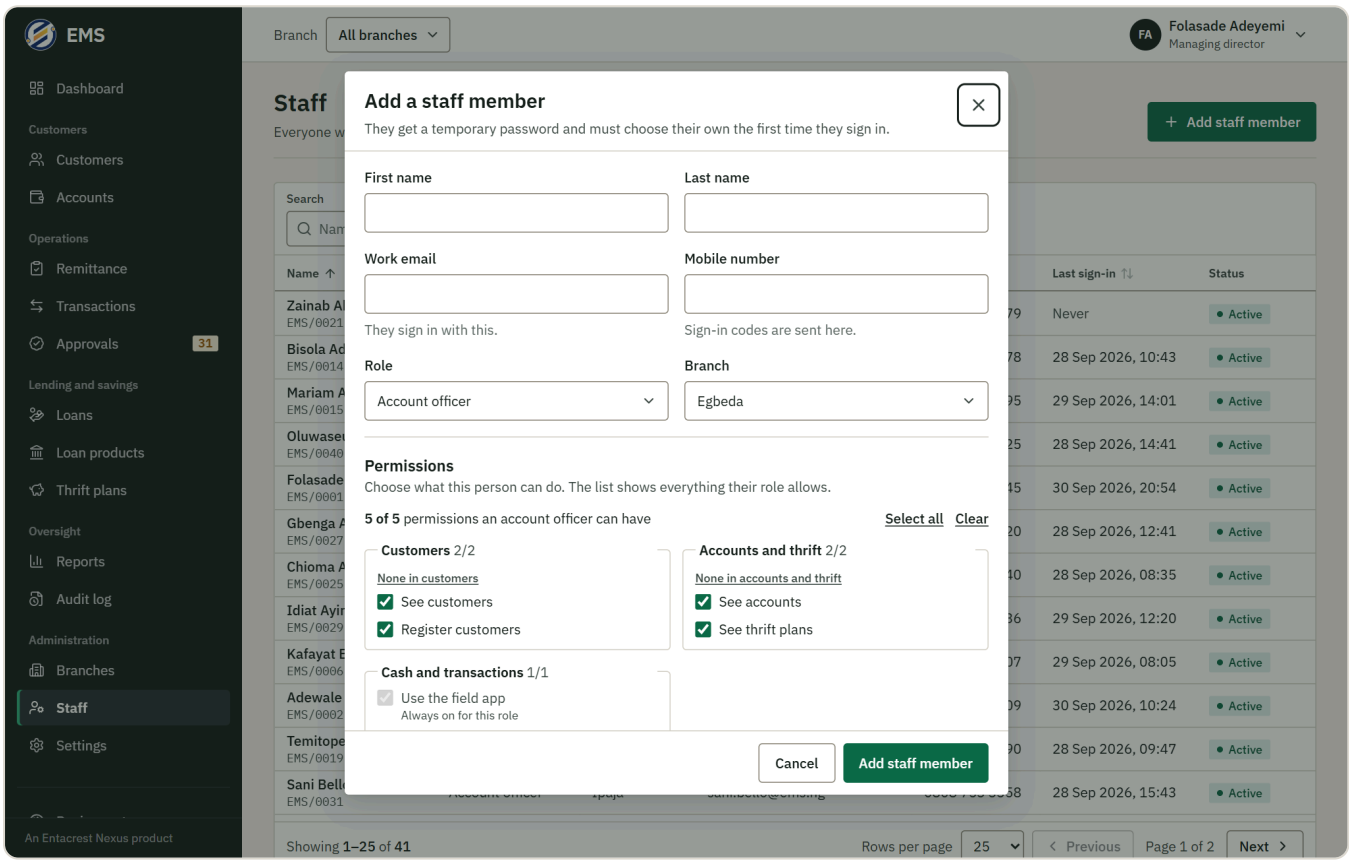
1. Go to **Administration** → **Branches** and click **“Add branch”**.
2. Enter the branch name, a two-digit **code**, address and phone. The code starts every account number in the branch, so it can’t change once accounts exist.
3. Add each **collection area** (e.g. Ipaja Market) and click **“Add area”**. Officers and customers are grouped by area.
4. Leave the manager as **“No manager yet”** for now. Pick them after you add staff.
5. Click **“Add branch”**. Your licence limits how many branches you can add.



Adding a branch with its collection areas.

Step 2 · Add your staff

1. Go to **Administration → Staff** and click **“Add staff member”**.
2. Enter their name, work email (they sign in with it) and mobile number (their codes go there).
3. Choose their **Role** and **Branch**.
4. Under **Permissions**, tick what this person may do. The list only shows what their role allows. Use **“Select all”** or **“Clear”** for speed.
5. Click **“Add staff member”**. EMS shows a **temporary password**, such as `Mango-River-4821`. Give it to them privately.



Adding a staff member and choosing their permissions.

> **Temporary passwords expire after 72 hours**
>
> The staff member must sign in and choose their own password within 3 days. If they miss it, open their record in Staff and reset the password to get a new one.

Then go back to **Branches** and choose each branch’s manager.

Step 3 · Create your products

Thrift plans

Lending and savings → **Thrift plans**. For each plan, set the contribution amount and how often it is paid (daily, weekly, monthly), the cycle length, the commission EMS takes, and whether customers may withdraw before the cycle ends.

Loan products

Lending and savings → **Loan products**. For each product, set:

- **Interest**: a rate **per month** or **for the whole loan** (overall).
- **Processing fee**: a percentage of the loan, a flat amount, or none.
- **Late payment fee**: a percentage or a flat amount, charged **daily** on overdue instalments.
- The amount range, tenor and repayment frequency, and whether it needs guarantors or collateral.

Step 4 · Check your settings

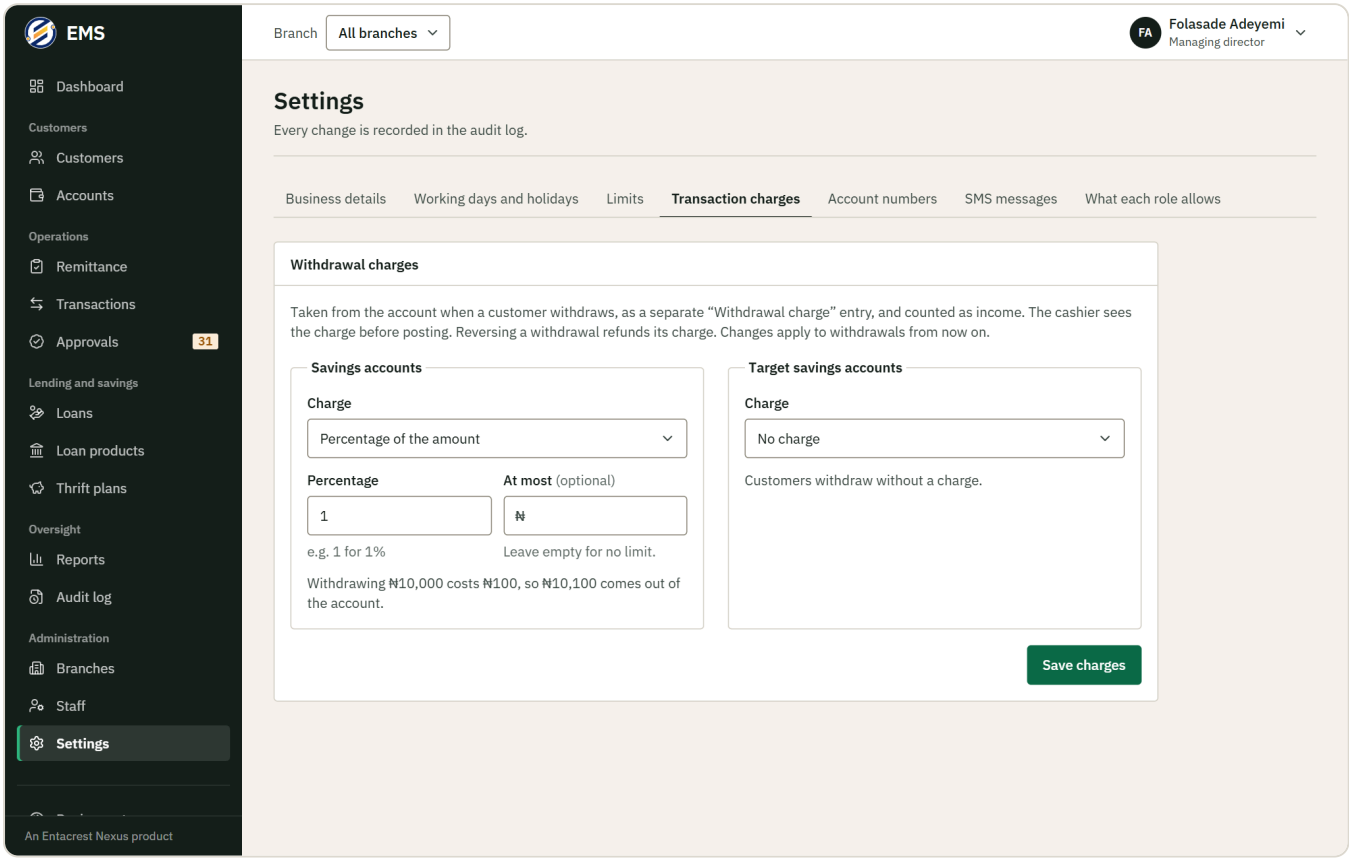
Go to **Administration** → **Settings**. Every change is recorded in the audit log.

Tab	What to set
Business details	Name, address and phone shown on reports and receipts.
Working days and holidays	Days you open, and public holidays. Repayment dates skip days you are closed.
Limits	Amounts above which withdrawals and transfers need a manager’s approval, and other limits.
Transaction charges	Withdrawal charges for savings and target savings (see below).
Account numbers	How account numbers are formed.
SMS messages	The wording of texts customers receive.
What each role allows	The most any person in each role may be given.

Transaction charges

Charge customers when they withdraw from a **savings** or **target savings** account.

1. Open the **“Transaction charges”** tab.
2. For each account type, pick **“No charge”**, **“Percentage of the amount”** (with an optional **At most** cap) or a flat amount.
3. Check the example under the fields, then click **“Save charges”**.



A 1% charge on savings withdrawals: withdrawing ₦10,000 takes ₦10,100 from the account.

The charge appears on the statement as a separate **Withdrawal charge** entry and counts as income. If a withdrawal is reversed, its charge is refunded. New charges apply to withdrawals from then on.

CHAPTER 7

Using EMS on several computers

All staff must work on the **same** data. Choose one of these set-ups. Only the head office computer needs a licence.

Option A · One office, shared over the office network

Best when everyone is in the same building on the same Wi-Fi or network.

On the head office computer (the one with the licence):

1. Open **File** → **Share on office network** and turn it on.
2. Open **File** → **Office network address…** and write down the address shown (for example `http://192.168.1.20:47821`).
3. If Windows Firewall asks, allow EMS on **private** networks.
4. Keep this computer switched on while staff work.

On each staff computer:

1. Install EMS (Chapter 3).
2. On the Activate screen, click **“Connect to your organisation”**. Do not enter a licence.
3. Choose **“On my organisation’s EMS server”**, enter the address from above and click **“Test connection”**.
4. Click **“Save and restart EMS”**, then sign in with your own email and password.

Option B · Several branches, using Supabase (online database)

Best when branches are in different places. Every computer connects to the same online database.

1. Create a free project at **supabase.com**. Choose a **database password** using letters and numbers only, and keep it safe.
2. In Supabase, click **“Connect”**, choose **Session pooler** and copy the connection string. Replace `[YOUR-PASSWORD]` with your database password (not your Supabase account password).
3. In EMS on the head office computer, open **File** → **Connection settings…** and choose **“In a Supabase (PostgreSQL) database”**.
4. Paste the string and click **“Test connection”**. To move your existing data, tick the option to copy this computer’s data into the new (empty) database.
5. Click **“Save and restart EMS”**. The first start creates the tables and can take a minute.

> **Keep the connection string secret**
> It contains your database password. Anyone who has it can read your data. If it is ever shared by mistake, reset the database password in Supabase.

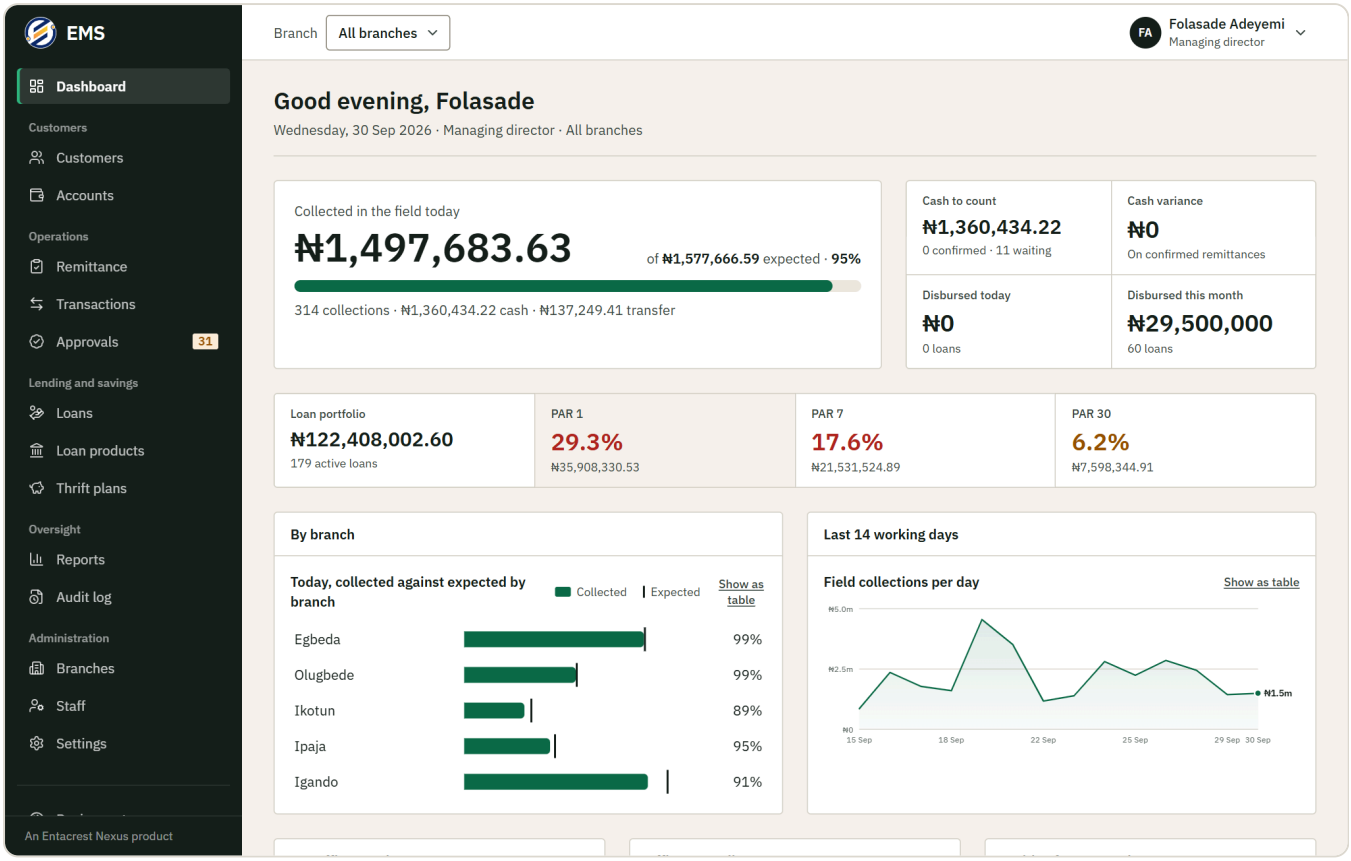
> **Only one EMS server may use a database at a time.**
> For many branches, the simplest set-up is to host EMS online (Option C) and let everyone connect to it.

Option C · Hosted online (web)

Entacrest can host EMS on the internet (for example on Render) with your Supabase database. Staff then open it in a browser, or connect the desktop app to it with **“On my organisation’s EMS server”** and the web address.

CHAPTER 8

Customers and accounts



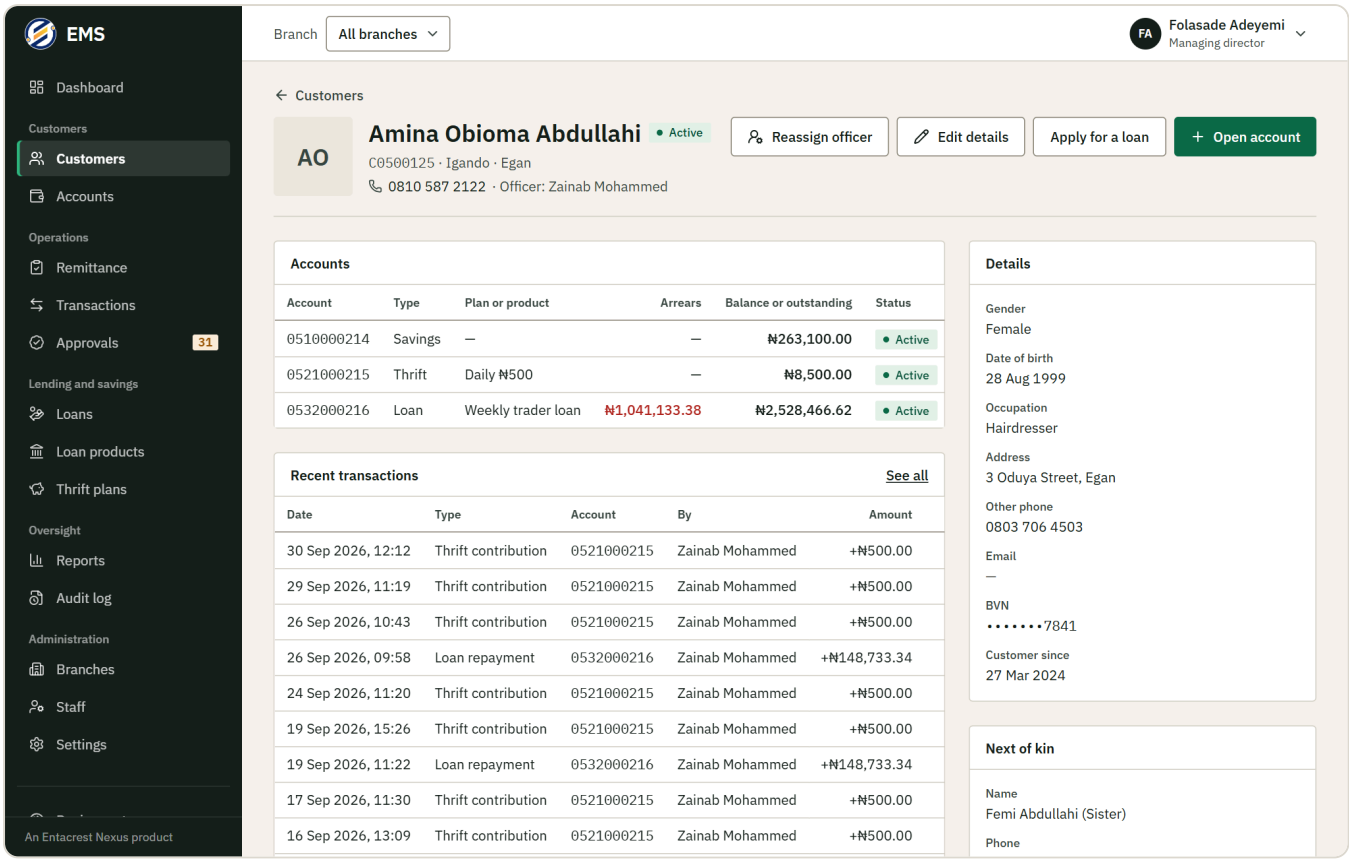
The Dashboard: today's collections, deposits, loans and income at a glance. Use the Branch picker at the top to switch branches.

Register a customer

1. Go to **Customers** and click **"Add customer"**. (Officers can also register customers in the field app.)
2. Enter their details: name, phone, address, area, ID (e.g. NIN or BVN), photo and next of kin.
3. Save. New customers may need a manager's approval before accounts are opened, depending on your set-up.

Open an account

1. On the customer's page, click **"Open account"** and choose **Savings**, **Target savings** or **Thrift** (choose the thrift plan).
2. The account number is created automatically. It starts with the branch code.



A customer’s page lists their accounts, loans and history.

Account page

Open any account to see its balance and statement. From here you can:

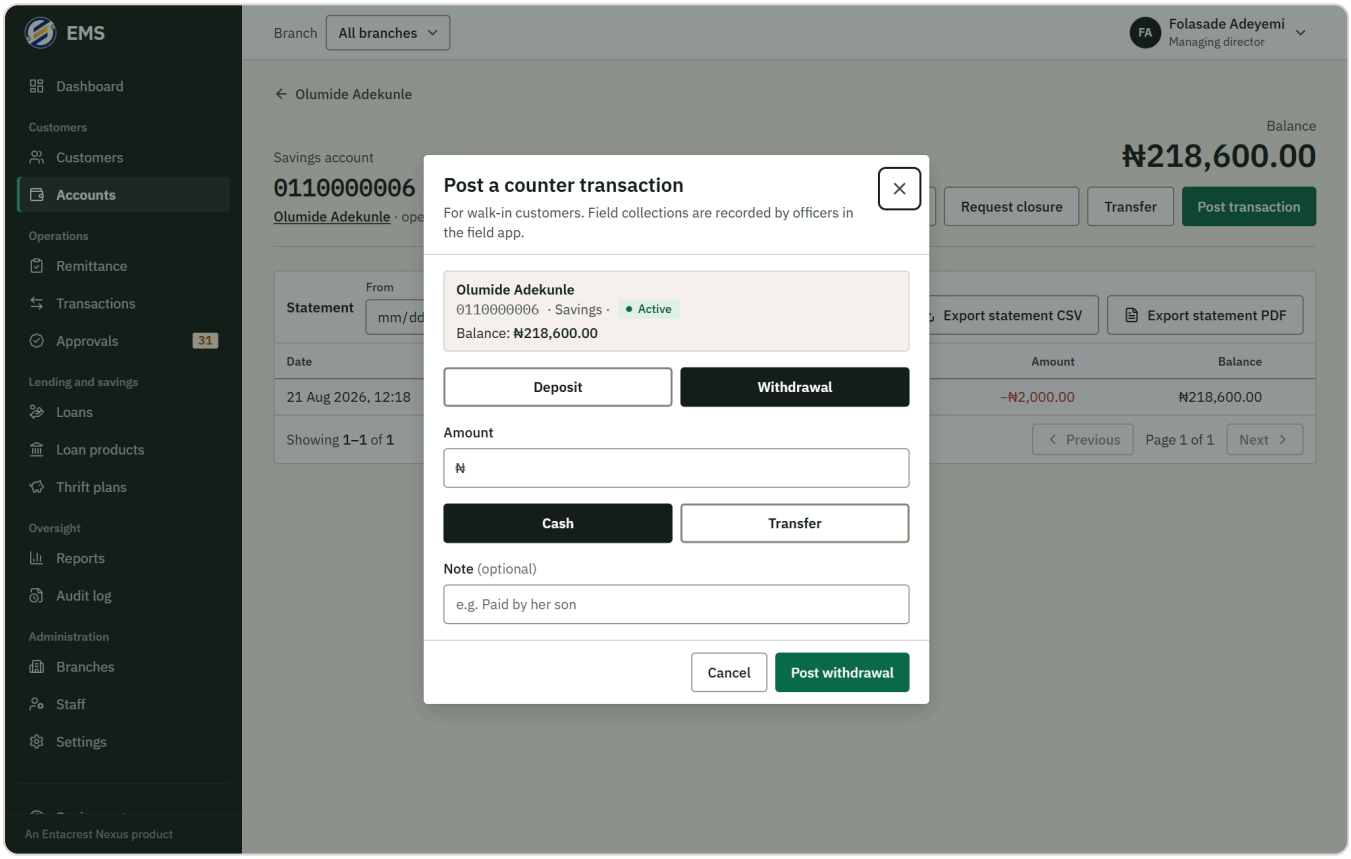
- **“Post transaction”**: deposit or withdraw at the counter (Chapter 9).
- **“Transfer”**: move money to another account (Chapter 10).
- **“Export statement PDF”** or **“CSV”**: choose dates first.
- **“Freeze”** the account, or **“Request closure”** (closure needs approval).

CHAPTER 9

Deposits, withdrawals and charges

For customers who come to the counter. Money collected in the market is recorded in the field app instead (Chapter 12).

1. Open the customer's account and click **"Post transaction"**. (Or use **Operations → Transactions → Post transaction** and search for the account.)
2. Choose **"Deposit"** or **"Withdrawal"**.
3. Enter the amount and choose **"Cash"** or **"Transfer"**. Add a note if useful.
4. For withdrawals, check the **charge** EMS shows before posting.
5. Click **"Post deposit"** or **"Post withdrawal"**. The customer gets an SMS receipt.



Posting a withdrawal at the counter.

> **Large withdrawals**
> If the amount is above the limit in Settings, it is not paid immediately. It goes to **Approvals** and is paid when a manager approves (Chapter 14).

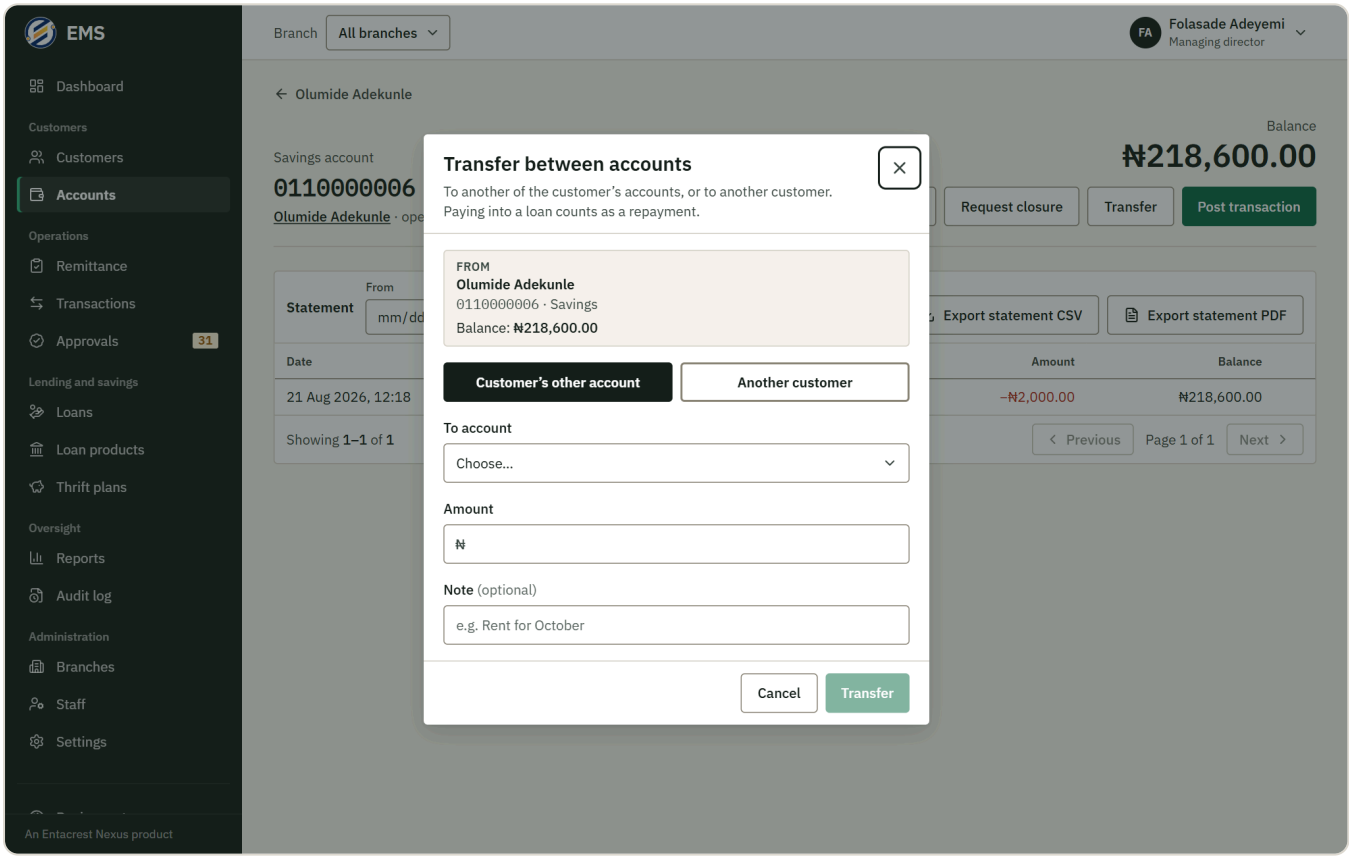
EMS won't let an account go below zero, including the charge.

CHAPTER 10

Transfers between accounts

Move money without cash changing hands: between a customer's own accounts, or from one customer to another.

1. Open the account the money comes **from** and click **"Transfer"**. (Or use **"Transfer"** on the Transactions page.)
2. Choose **"Customer's other account"** and pick it from the list, **or** choose **"Another customer"** and type their account number. EMS shows their name so you can check it's the right person.
3. Enter the amount and a note (e.g. "Rent for October").
4. Click **"Transfer"**.



Transfer between accounts.

- Both sides appear on their statements: **Transfer out** and **Transfer in**.
- Transferring **into a loan account** is recorded as a **loan repayment**. You can't pay more than the loan still owes.
- Transfers above the limit go to **Approvals** first.
- Frozen or closed accounts can't send or receive.
- Reversing a transfer reverses both sides together.

CHAPTER 11

Thrift (ajo/esusu) savings

Customers on a thrift plan contribute a fixed amount on a schedule, usually collected by an officer in the field. EMS takes the plan's commission each cycle.

Withdrawing from a thrift account

1. Open the thrift account and click **"Post transaction"**, then **"Withdrawal"**.
2. **Cycle complete:** withdrawing the full balance is recorded as a **thrift payout**, the customer gets an SMS, and a new cycle starts.
3. **Cycle not complete:** allowed only if the plan permits early withdrawals. Otherwise EMS explains that the plan doesn't allow withdrawals before the cycle ends.

Thrift withdrawals have no transaction charge. Large ones still need approval.

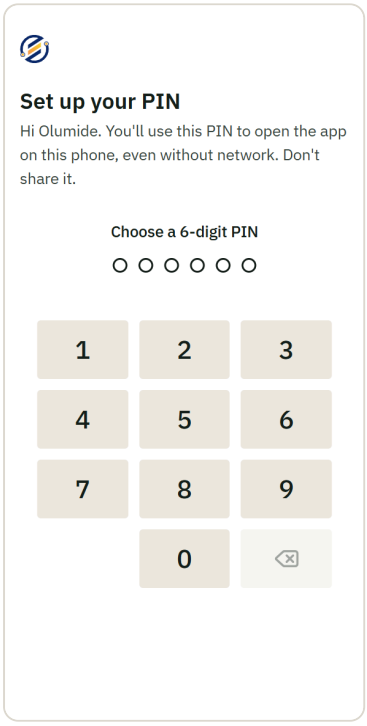
CHAPTER 12

Field collections and remittance

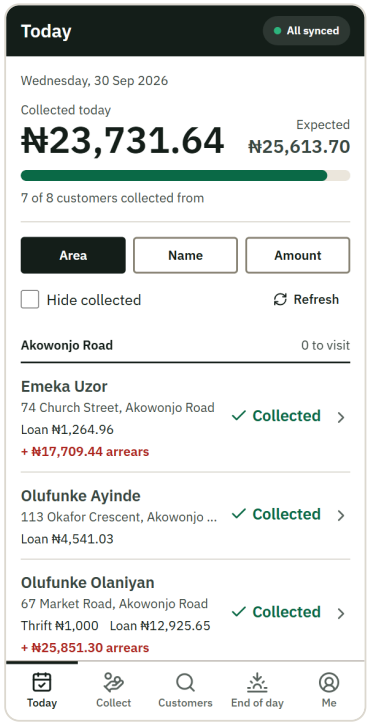
For account officers: the field app

Officers use a phone. In the desktop app it opens from **File → Field app**; online, it is the `/field` page of your EMS address.

1. Sign in with email, password and the SMS code.
2. The first time, choose a **6-digit PIN** and type it again. Afterwards, the PIN is all you need to open the app.
3. **"Today"** lists the customers to visit, grouped by area, with what each should pay and any arrears.
4. Tap a customer (or use **"Collect"**), enter the amount and save. The customer gets an SMS.
5. No network? Keep collecting. Entries are saved on the phone and sent when the network returns. The top badge shows **"All synced"** when everything is sent.
6. At the end of the day, open **"End of day"**, check the totals and submit. Then hand the cash to the accountant.



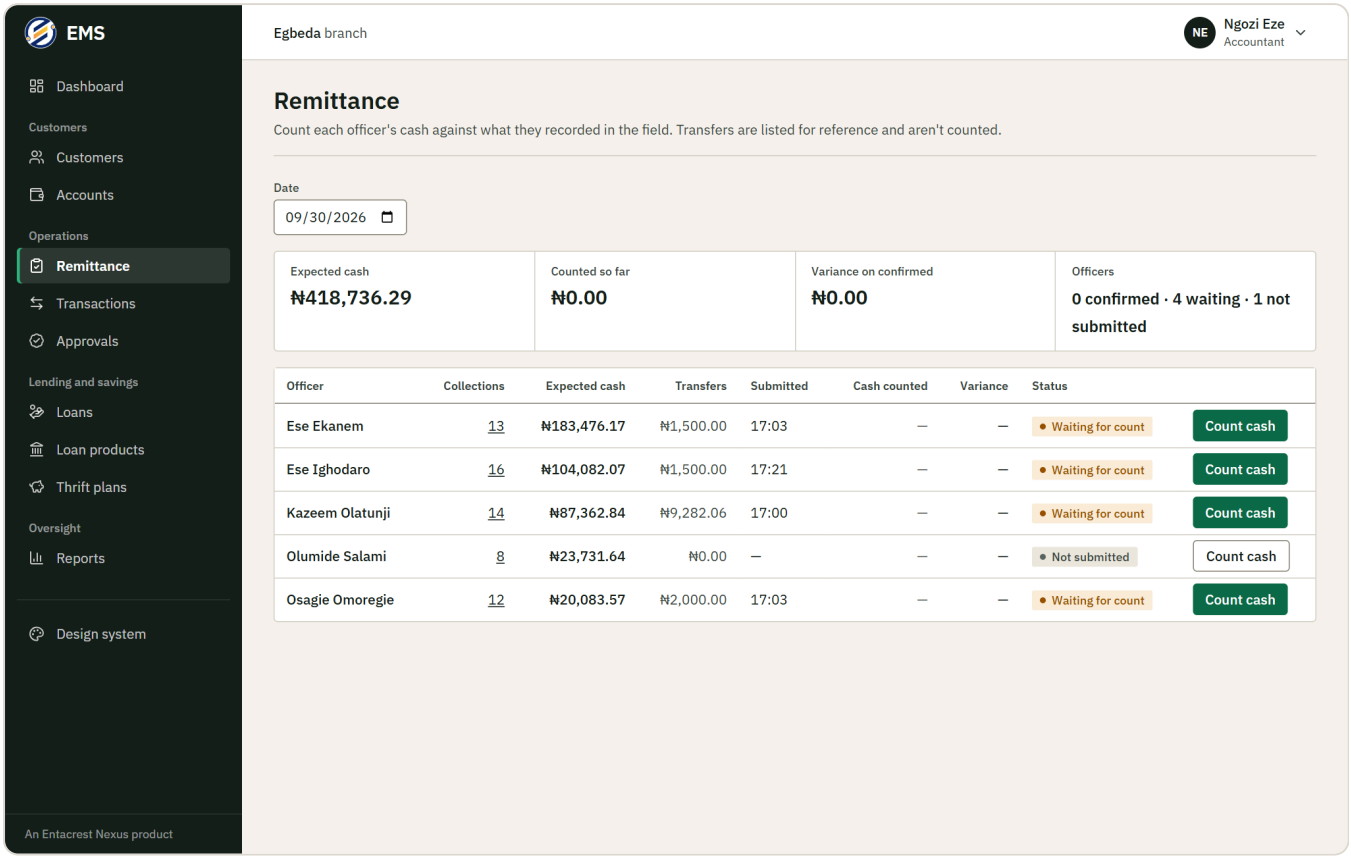
Setting up your 6-digit PIN.



Today: customers to visit and what's been collected.

For the accountant: remittance

1. Go to **Operations → Remittance** and pick the date.
2. Each officer shows the cash they are expected to hand in. Transfers are listed but not counted.
3. Count the officer's cash and click **"Count cash"**. Enter what you counted.
4. EMS shows any **variance** (shortage or excess). Confirm it.



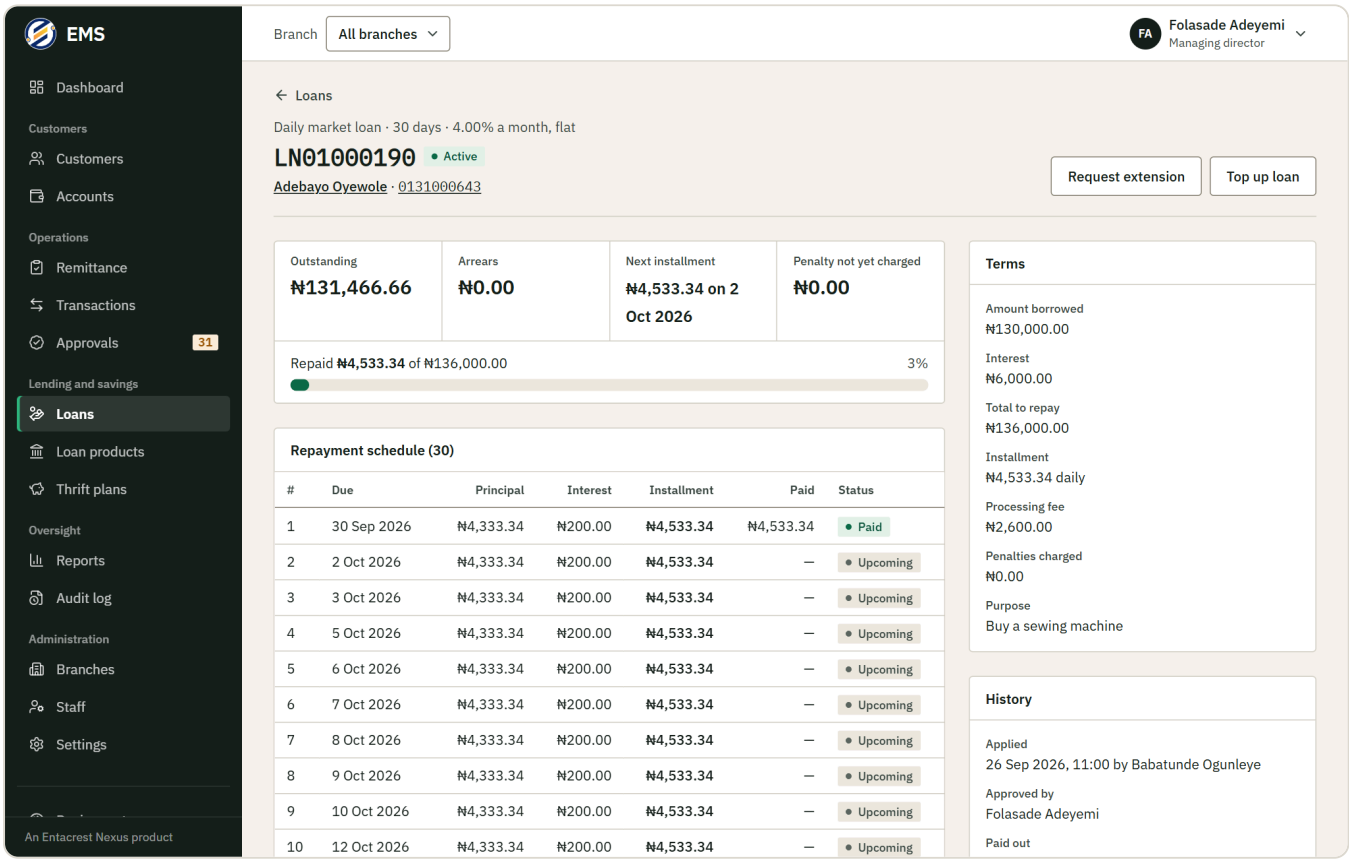
Remittance: counting each officer's cash.

CHAPTER 13

Loans

The life of a loan

1. **Apply.** Open the customer's page and click **"Apply for a loan"**. Choose the product, amount, tenor and purpose. Add guarantors or collateral if the product needs them. EMS shows the repayment schedule before you submit.
2. **Approve.** The application goes to **Approvals**. A manager approves or rejects it (Chapter 14).
3. **Pay out.** Open the loan under **Lending and savings** → **Loans** and click **"Pay out loan"**. The processing fee is taken as set on the product. For buy-now-pay-later products, use **"Pay supplier"** instead.
4. **Repay.** Repayments come from field collections, the counter, or transfers from the customer's savings. Each repayment settles the oldest instalment first.
5. **Late fees.** Overdue instalments attract the product's daily late fee. It shows as **Penalty**.
6. **Changes.** Use **"Request extension"** to give more time (needs approval), or **"Top up loan"** to lend more.



A loan: outstanding balance, arrears, repayment schedule and terms.

CHAPTER 14

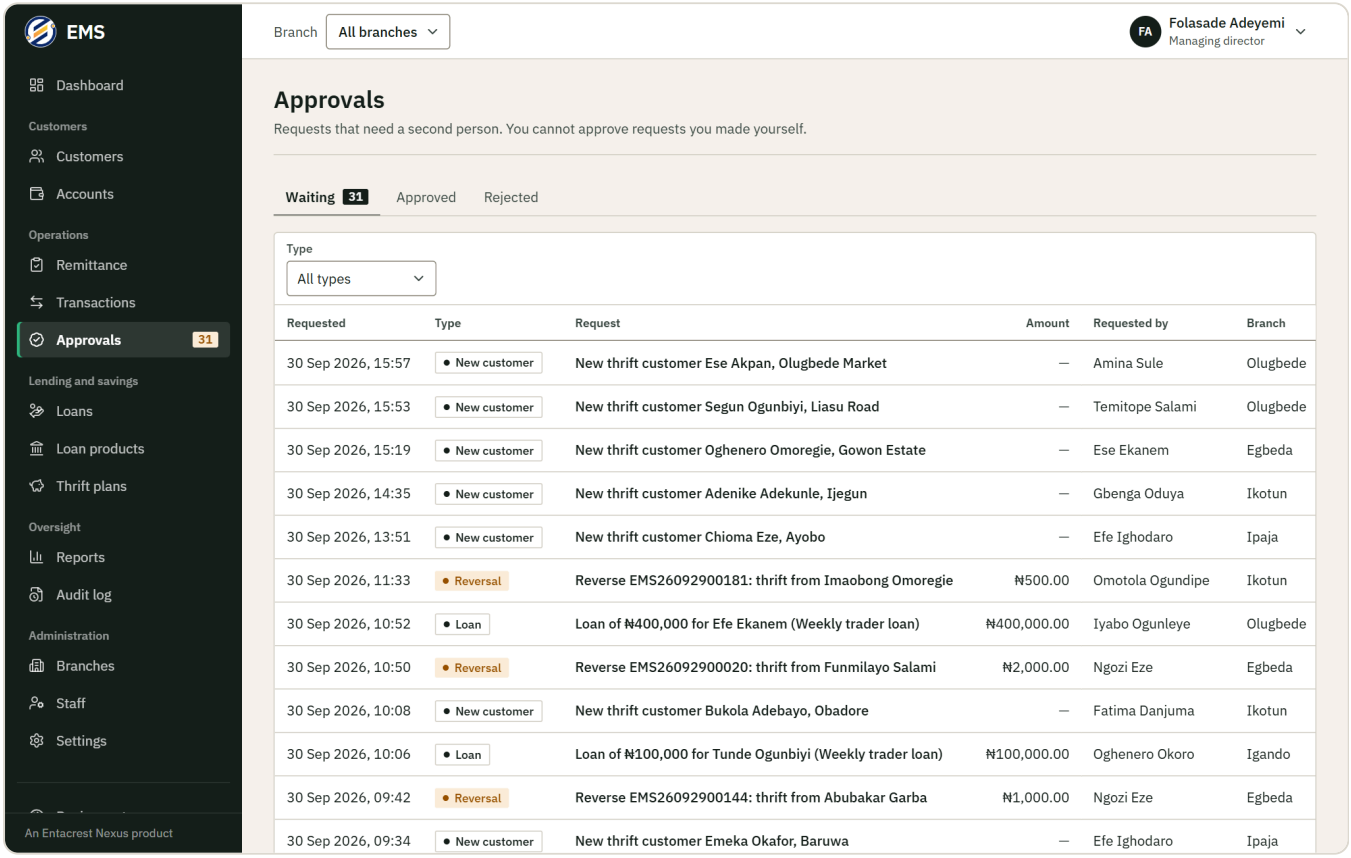
Approvals

Sensitive actions need a second person. The number next to **Approvals** in the menu shows how many are waiting.

These go to approval: new customer registrations, loan applications, loan extensions, large withdrawals, large transfers, reversals and account closures.

1. Go to **Operations → Approvals**.
2. Open a request and read the details and the reason given.
3. Click **Approve** or **Reject**, with a note if needed. Nothing changes until it is approved.

> **You can't approve your own request.**
> Someone else with approval permission must do it.



Approvals waiting for a decision.

CHAPTER 15

Correcting mistakes

Reverse a transaction

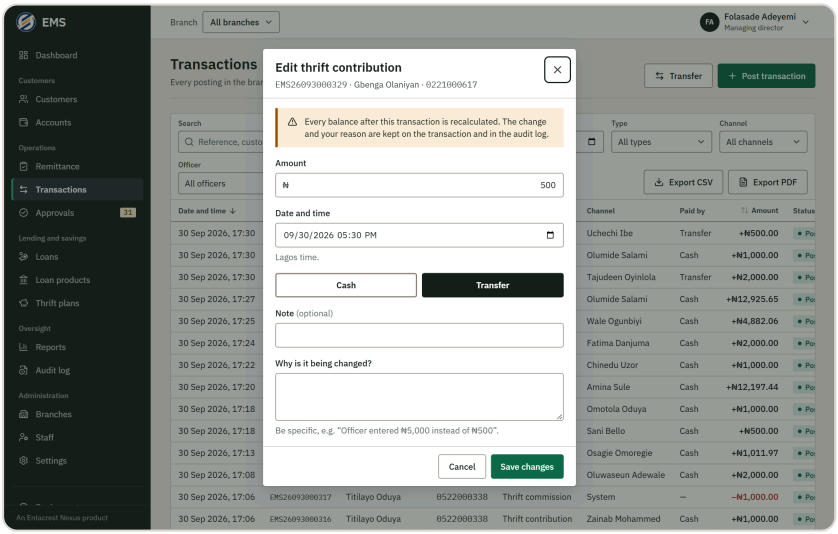
Use this when a transaction should never have happened (wrong customer, duplicate).

1. Open the transaction (from Transactions or a statement) and click **“Request reversal”**.
2. Explain why. Be specific: the approving manager reads it.
3. A manager approves it. The money goes back, charges are refunded, and both sides of a transfer are reversed.

Edit a transaction

Use this when a transaction was real but recorded wrongly (wrong amount, date, payment method or note). **Only the MD, general managers and branch managers can edit.** Branch managers can edit their own branch only.

1. Open the transaction and click **“Edit”**.
2. Change the amount, date and time, Cash/Transfer or note.
3. In **“Why is it being changed?”**, give a clear reason, e.g. “Officer entered ₦5,000 instead of ₦500”.
4. Click **“Save changes”**.



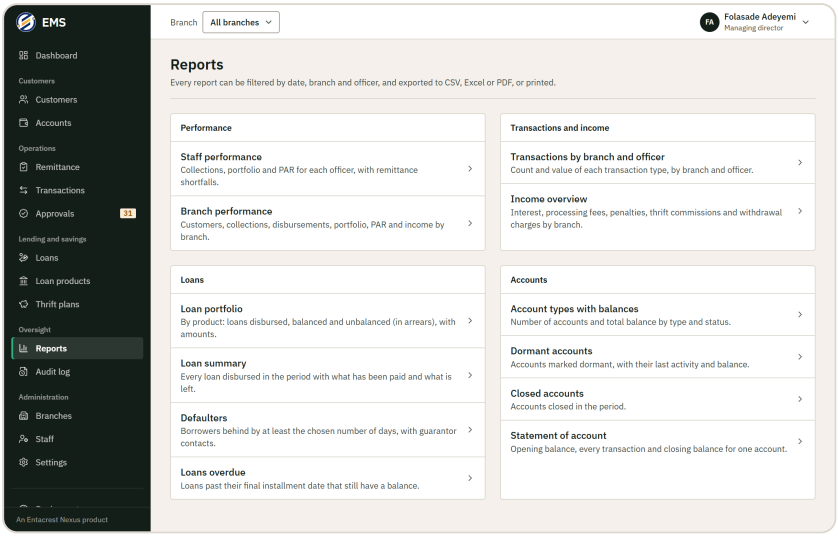
Editing a transaction.

- Every balance after it is recalculated. Editing a loan repayment updates what the loan still owes.
- The transaction is tagged **Edited** and shows a **Changes** history. The edit is also in the audit log.
- EMS refuses edits that would leave an account short, dates in the future, or amounts on loan payouts (only their date and note can change).

CHAPTER 16

Reports and the audit log

Go to **Oversight** → **Reports**, choose a report, set the dates and branch, then view it or export it to PDF or CSV.



The Reports page.

Report	Shows
Staff performance	What each officer collected against target.
Branch performance	Branches compared.
Transactions by branch and officer	Every posting, grouped.
Income overview	Interest, fees, penalties, thrift commission and withdrawal charges.
Loan portfolio / Loan summary	Money out, repaid and outstanding.
Defaulters / Loans overdue	Who is behind and by how much.
Account types with balances	Total held in each account type.
Dormant accounts / Closed accounts	Accounts with no recent activity, or closed.
Statement of account	One account’s statement for any period.

Audit log

Oversight → **Audit log** records who did what and when: sign-ins, postings, approvals, edits and settings changes. Nobody can delete entries.

CHAPTER 17

Practising with the demo workspace

The desktop app includes a **demo workspace** with sample branches, staff, customers and loans. It is completely separate from your real data, so it is ideal for training new staff.

1. Open **File → Demo workspace → Demo back office** (or **"Demo field app"**).
2. Sign in with any demo staff email, e.g. `md@ems.ng`, `manager@ems.ng`, `cashier@ems.ng`, `accountant@ems.ng` or `officer@ems.ng`.
3. Password: `Demo@2026`. Sign-in code: `123456`.
4. To start again with fresh sample data, use **File → Demo workspace → Reset demo data...**

CHAPTER 18

Backups

If your data is on this computer, back it up at least weekly, ideally daily.

1. Open **File → Back up live data...**
2. Choose where to save it: a USB drive, or a Google Drive / OneDrive folder.
3. Keep several copies from different dates.

> **Don't run EMS from a cloud folder**
> Google Drive and OneDrive are fine for storing backup copies, but EMS's live database must not be kept in them. To share data between computers, use Chapter 7.

If you use Supabase, it keeps its own backups. **File → Open data folder** shows where local data is kept.

CHAPTER 19

Troubleshooting

Problem	What to do
I didn’t get my sign-in code	If SMS isn’t connected yet, the code is in **File → SMS outbox** on the head office computer. Codes expire after 5 minutes; sign in again for a new one. Check the mobile number on your staff record.
I forgot my password	Ask the MD or general manager to reset it in **Staff**. You’ll get a new temporary password.
My temporary password doesn’t work	It expires after 72 hours. Ask for it to be reset.
A new computer shows the licence screen	Staff computers don’t need a licence. Click **“Connect to your organisation”** (Chapter 7).
“Can’t reach the server” on a staff computer	Check the head office computer is on, EMS is open and **File → Share on office network** is on, and both computers are on the same network. Check the address has not changed.
“Another EMS server is using this database”	Only one EMS server may use a Supabase database. Close EMS on the other computer, or connect that computer to the organisation’s server instead.
Stuck on “Checking the connection…”	Use the **Session pooler** string from Supabase (port 5432), with your database password, not your account password.
The licence has expired	Contact Entacrest to renew. **Help → Licence details** shows the expiry date.
A withdrawal or transfer didn’t go through	If it was large, it is waiting in **Approvals**. If the account is frozen, closed or short of funds, EMS says so.
I can’t see a menu or button	Your role or permissions don’t include it. Ask the MD.
Windows says “Windows protected your PC”	Click **“More info” → “Run anyway”**.

Still stuck? Contact Entacrest support, and include a screenshot of the message you see.

CHAPTER 20

Roles at a glance

These are typical. The MD chooses each person’s exact permissions when adding them, within what their role allows.

Role	Usually does	Edit transactions
Managing director (MD)	Everything, across all branches. Sets up the business, staff and settings.	Yes
General manager	Runs all branches: approvals, reports, staff.	Yes
Branch manager	Runs one branch: takes and approves loan applications, approves requests and reversals; reviews reports.	Own branch
Administrator	Branches, staff and settings.	No
Accountant	Remittance, cash counts, reports.	No
Cashier	Counter deposits, withdrawals and transfers; pays out approved loans; requests reversals.	No
Account officer	Registers customers, collects in the field app and follows up loans.	No

Good practice

Give each person only what their job needs. Everyone signs in as themselves, and never shares passwords or PINs, so the audit log shows who did what.